

4 b) $f(m,n) = m^2 n^2$

a) Solution: Let, $f(m,n) = -1 \in \mathbb{Z}$.

Sc We know that $\forall m, n \in \mathbb{Z}, m^2 \geq 0$ and $n^2 \geq 0$
 $\therefore m^2 n^2 \geq 0$. There does not exist $m, n \in \mathbb{Z}$ s.t. $f(m,n) = -1$

$\therefore$ By Defn. 7, f is not onto Π

b) c) $f(m,n) = m$

S Solution: This is very easy to show that f is onto.

d) $f(m,n) = |n|$

E Solution: Let, $f(m,n) = x$, where $x \in \mathbb{Z}$.

T $\therefore |n| = x$ If $x < 0$, then n does not exist

C $\therefore \forall n \in \mathbb{Z}, |n| \geq 0$. By Defn. 7, f is ^{not} onto Π

e) $f(m,n) = m - n$

S Solution: Let, $f(m,n) = x$, where $x \in \mathbb{Z}$.

C $\therefore m - n = x$. Let, $m = x + 1, n = 1 \therefore m - n = x$

$\therefore x \in \mathbb{Z}, x + 1 \in \mathbb{Z}, 1 \in \mathbb{Z} \therefore \forall x \in \mathbb{Z}, \exists m, n \in \mathbb{Z}$ s.t. $f(m,n) = x$.

By Defn. 7, f is onto Π

20) Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$ that is

a) one-to-one, but not onto.

Solution: $f(n) = n^2$.

Let, $x, y \in \mathbb{N}$ and $f(x) = f(y) \Rightarrow x^2 = y^2 \Rightarrow x^2 - y^2 = 0 \Rightarrow (x-y)(x+y) = 0$.

Either $(x-y) = 0$ or $(x+y) = 0$. We know that $x \geq 0$ and $y \geq 0, \therefore x, y \in \mathbb{N}$.

$\therefore (x+y) \geq 0, (x+y) = 0$ iff $x = y = 0$.

$x - y = 0 \Rightarrow x = y$. By Defn. 5, f is one-to-one from $\mathbb{N}$ to $\mathbb{N}$.

Let $f(x) = 3$. We know $3 \in \mathbb{N}$. $\therefore x^2 = 3 \Rightarrow x = \pm\sqrt{3} \notin \mathbb{N}$.

$\therefore$ There does not exist $x \in \mathbb{N}$ s.t. $f(x) = 3$. By Defn. 7, f is not onto. Π